

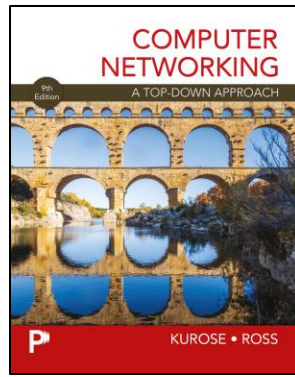
CSC 361

Computer Networks

Network Layer

(Chapter 5 – Control Plane)

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Summer 2026



Network Layer: “Control Plane” Roadmap (1 of 10)

- introduction
- routing protocols
 - link state
 - distance vector
- intra-ISP routing: OSPF
- routing among ISPs: BGP
- SDN control plane
- Internet Control Message Protocol

Network-Layer Functions

- **forwarding**: move packets from router's input to appropriate router output

data plane

- **routing**: determine route taken by packets from source to destination

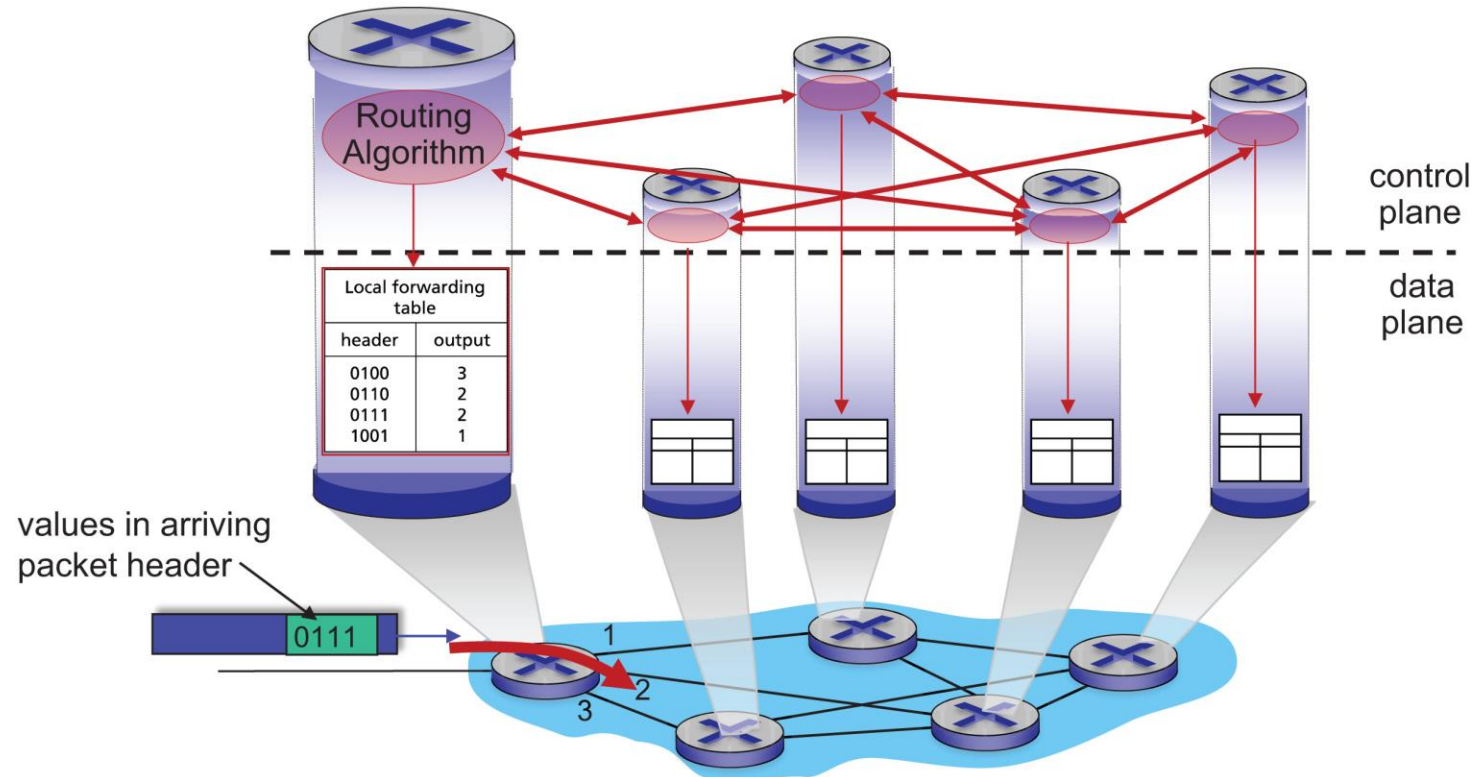
control plane

Two approaches to structuring network control plane:

- per-router control (traditional)
- logically centralized control (software defined networking)

Per-router Control Plane (1 of 2)

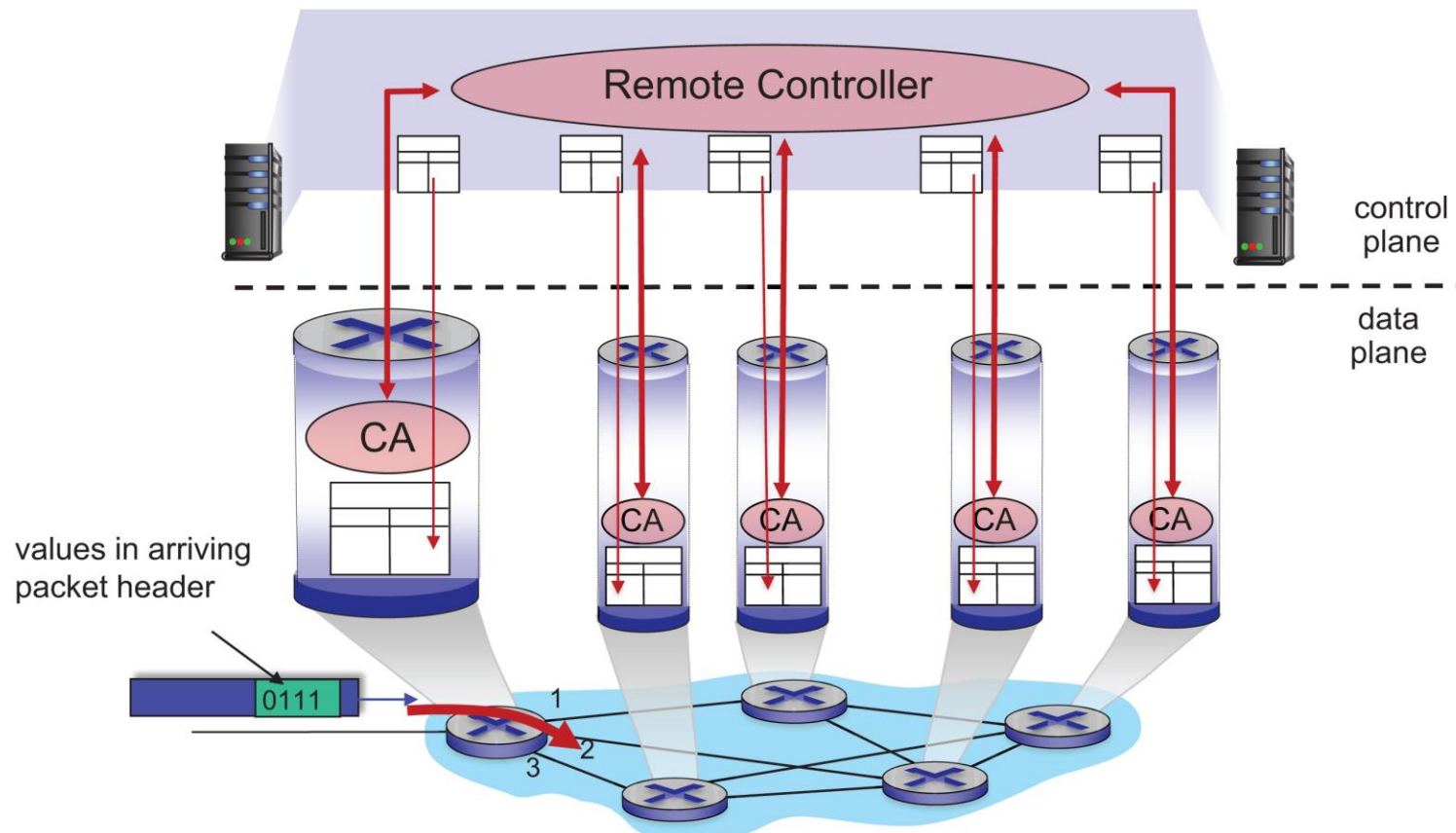
Individual routing algorithm components **in each and every router** interact in the control plane



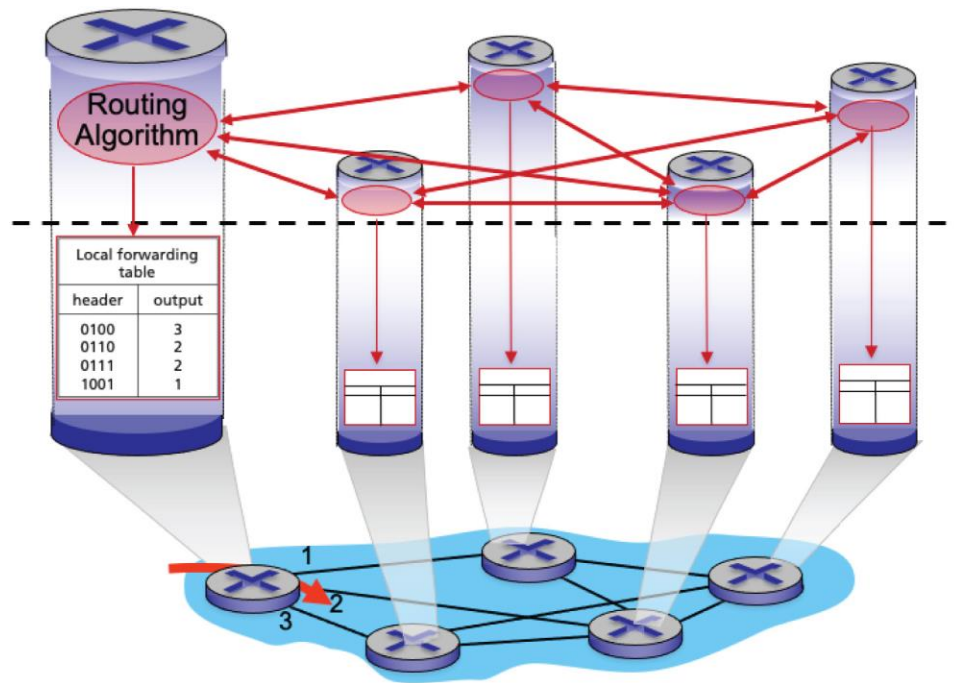
Software-Defined Networking (SDN) Control Plane

(1 of 2)

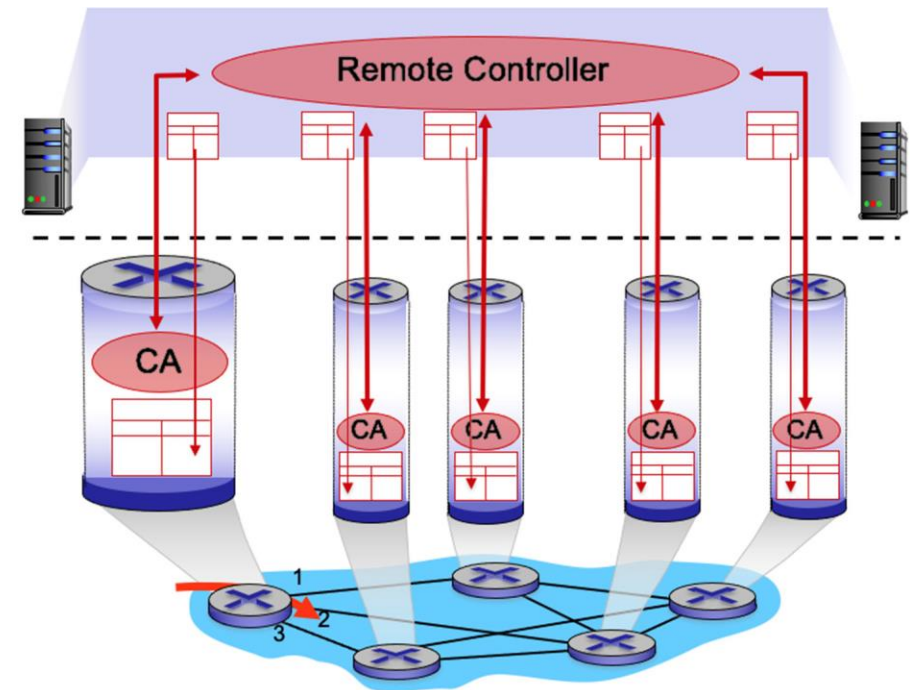
Remote controller computes, installs forwarding tables in routers



Per-router control plane



SDN control plane



Network Layer: “Control Plane” Roadmap (2 of 10)

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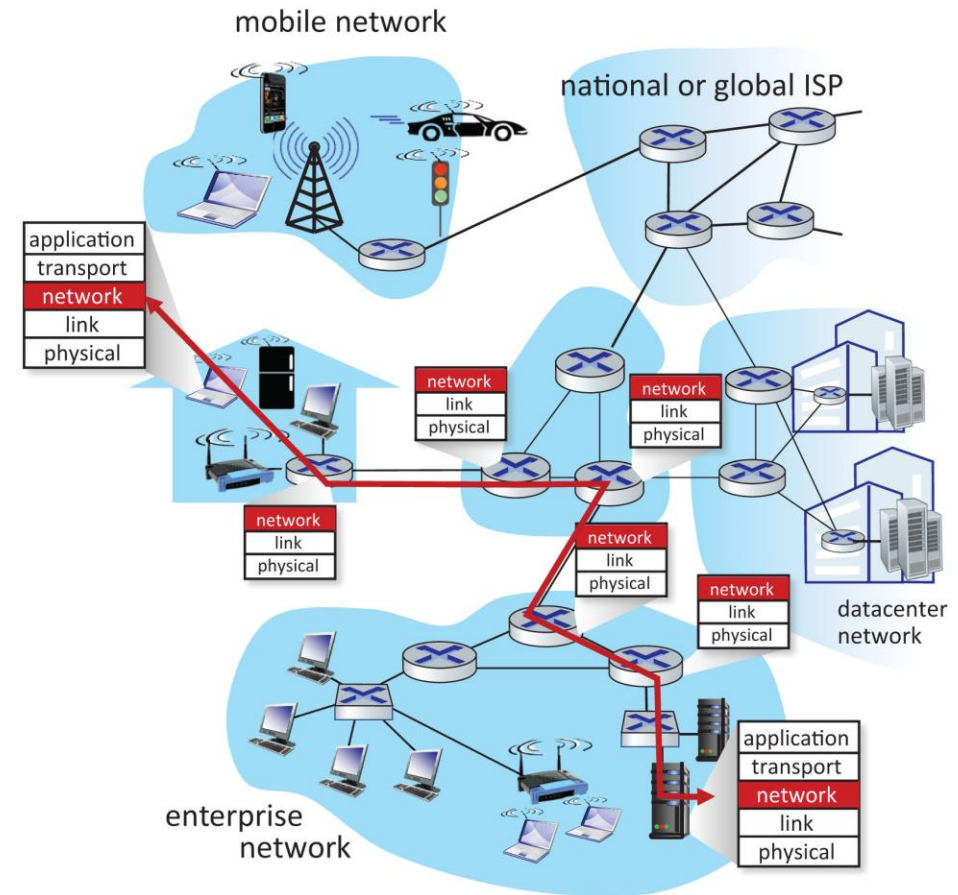


- network management, configuration
 - SNMP
 - NETCONF/YANG

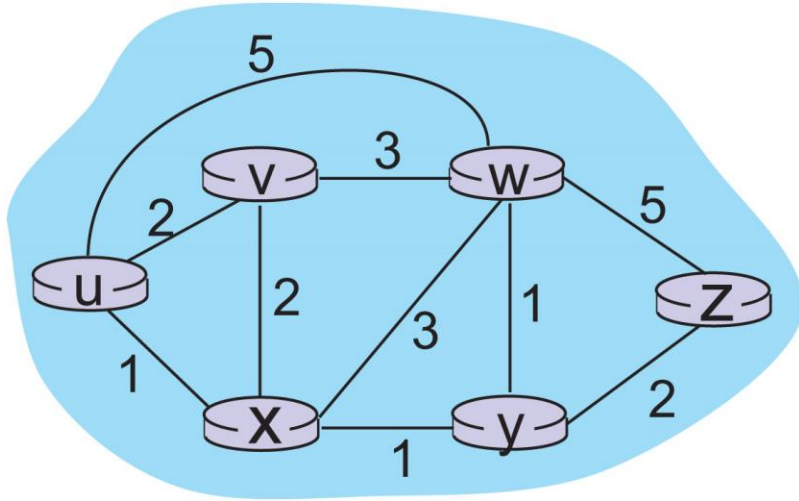
Routing Protocols

Routing protocol goal: determine “good” paths (equivalently, routes), from sending hosts to receiving host, through network of routers

- **path:** sequence of routers packets traverse from given initial source host to final destination host
- **“good”:** least “cost”, “fastest”, “least congested”
- routing: a “top-10” networking challenge!



Graph Abstraction: Link Costs



$c_{a,b}$: cost of **direct** link connecting a and b

e.g., $c_{w,z} = 5, c_{u,z} = \infty$

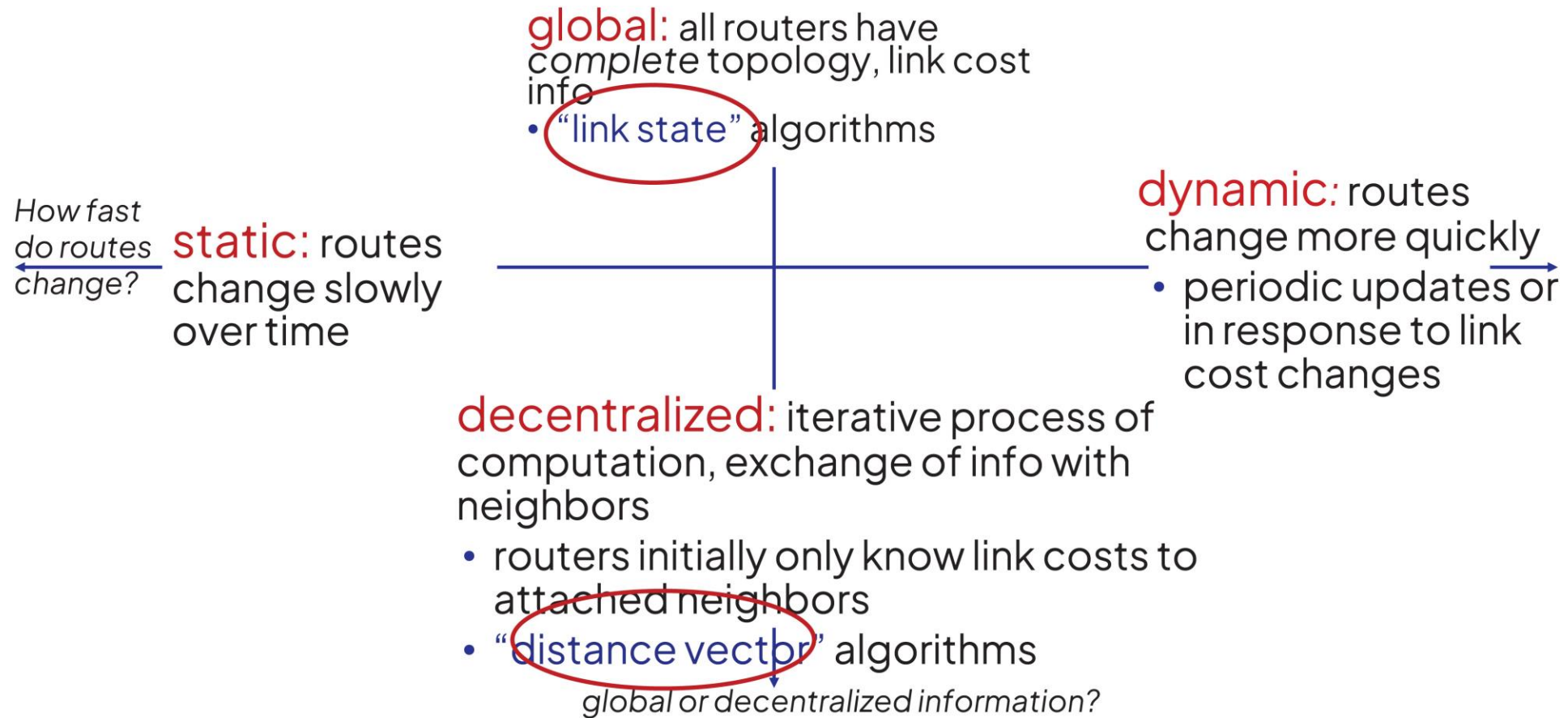
cost defined by network operator:
could always be 1, or inversely related
to bandwidth, or inversely related to
congestion

graph: $G = (N, E)$

N : set of routers $= \{u, v, w, x, y, z\}$

E : set of links $= \{ (u, v), (u, x), (v, x), (v, w), (x, w), (x, y), (w, y), (w, z), (y, z) \}$

Routing Algorithm Classification



Network Layer: “Control Plane” Roadmap (3 of 10)

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Dijkstra's Link-State Routing Algorithm (1 of 2)

- **centralized**: network topology, link costs known to **all** nodes
 - accomplished via “link state broadcast”
 - all nodes have same info
- computes least cost paths from one node (“source”) to all other nodes
 - gives **forwarding table** for that node
- **iterative**: after k iterations, know least cost path to k destinations

notation

- $c_{x,y}$: **direct** link cost from node x to y ; $= \infty$ if not direct neighbors
- $D(v)$: **current** estimate of cost of least-cost-path from source to destination v
- $p(v)$: predecessor node along path from source to v
- N' : set of nodes whose least-cost-path **definitively** known

Dijkstra's Link-State Routing Algorithm (2 of 2)

1 *Initialization:*

2 $N' = \{u\}$ */* compute least cost path from u to all other nodes */*

3 for all nodes v

4 if v adjacent to u */* u initially knows direct-path-cost only to direct neighbors */*

5 then $D(v) = c_{u,v}$ */* but may not be minimum cost! */*

6 else $D(v) = \infty$

7

8 *Loop*

9 find w not in N' such that $D(w)$ is a minimum

10 add w to N'

11. update $D(v)$ for all v adjacent to w and not in N' :

12 **$D(v) = \min (D(v), D(w) + c_{w,v})$**

13 */* new least-path-cost to v is either old least-cost-path to v or known*

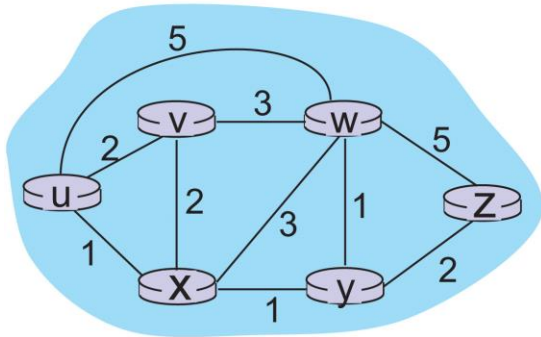
14 *least-cost-path to w plus direct-cost from w to v */*

15 *until all nodes in N'*

Dijkstra's Algorithm: An Example (1 of 13)

Step	N'	^v D(v),p(v)	^w D(w),p(w)	^x D(x),p(x)	^y D(y),p(y)	^z D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1						
2						
3						
4						
5						

D(w),p(w)
5,u
4,x
3,y
3,y



Initialization (step 0):

For all *a*: if *a* adjacent to *u* then $D(a)=c_{u,a}$

Dijkstra's Algorithm: An Example (2 of 13)

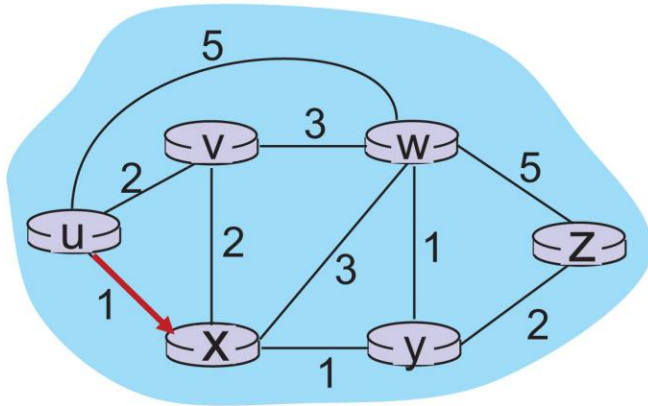
Step	N'	$D(v), p(v)$	$D(w), p(w)$	$D(x), p(x)$	$D(y), p(y)$	$D(z), p(z)$
0	u	2, u	5, u	1, u	∞	∞
1	u, x					
2						
3						
4						
5						

$D(w), p(w)$
 5, u
 4, x
 3, y
 3, y

8 Loop

9 find a not in N' such that $D(a)$ is a minimum

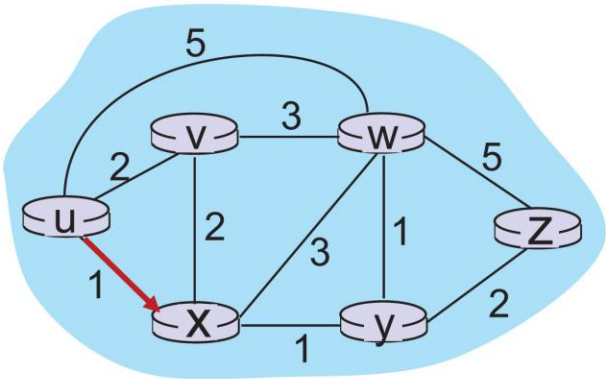
10 add a to N'



Dijkstra's Algorithm: An Example (3 of 13)

Step	N'	V D(v),p(v)	W D(w),p(w)	X D(x),p(x)	Y D(y),p(y)	Z D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2						
3						
4						
5						

D(w),p(w)
5,u
4,x
3,y
3,y



8 Loop

- 9 find a not in N' such that $D(a)$ is a minimum
- 10 add a to N'
- 11 update $D(b)$ for all b adjacent to a and not in N' :

$$D(b) = \min (D(b), D(a) + c_{a,b})$$

$$D(v) = \min (D(v), D(x) + c_{x,v}) = \min (2, 1+2) = 2$$

$$D(w) = \min (D(w), D(x) + c_{x,w}) = \min (5, 1+3) = 4$$

$$D(y) = \min (D(y), D(x) + c_{x,y}) = \min (\infty, 1+1) = 2$$

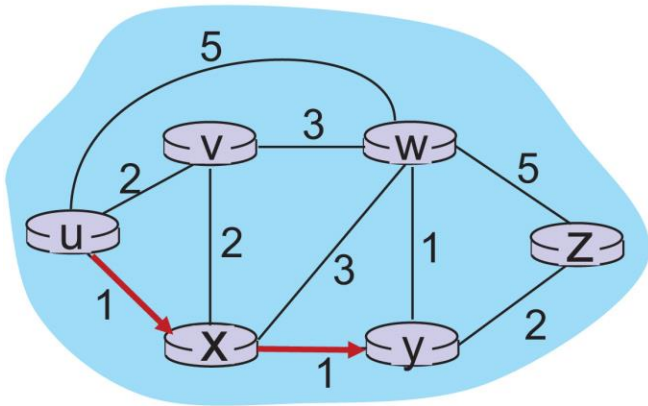


Dijkstra's Algorithm: An Example (4 of 13)

Step	N'	v D(v),p(v)	w D(w),p(w)	x D(x),p(x)	y D(y),p(y)	z D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2	uxy					
3						
4						
5						

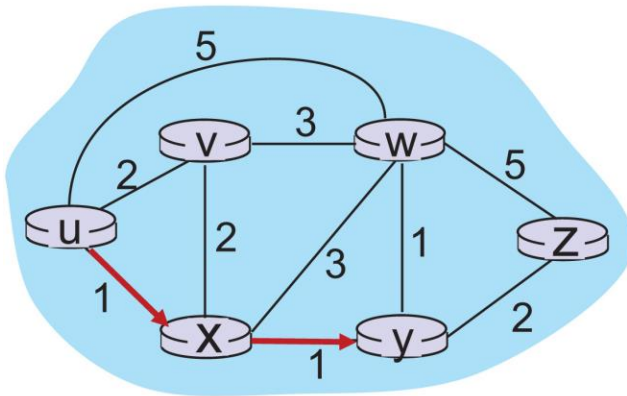
D(w),p(w)
5,u
4,x
3,y
3,y

- 8 Loop
- 9 find *a* not in *N'* such that *D(a)* is a
- 10 minimum
- add *a* to *N'*



Dijkstra's Algorithm: An Example (5 of 13)

		v	w	x	y	z
Step	N'	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2	uxy	2,u	3,y			4,y
3						
4						
5						



8 Loop

9 find a not in N' such that $D(a)$ is a minimum

10 add a to N'

11 update $D(b)$ for all b adjacent to a and not in N' :

$$D(b) = \min (D(b), D(a) + c_{a,b})$$

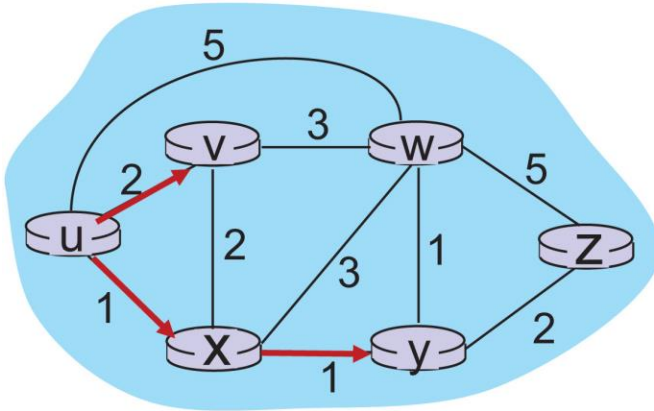
$$D(w) = \min (D(w), D(y) + c_{y,w}) = \min (4, 2+1) = 3$$

$$D(z) = \min (D(z), D(y) + c_{y,z}) = \min (\infty, 2+2) = 4$$



Dijkstra's Algorithm: An Example (6 of 13)

Step	N'	$D(v), p(v)$	$D(w), p(w)$	$D(x), p(x)$	$D(y), p(y)$	$D(z), p(z)$
0	u	2, u	5, u	1, u	∞	∞
1	ux	2, u	4, x		2, x	∞
2	uxy	2, u	3, y			4, y
3	uxyv					
4						
5						



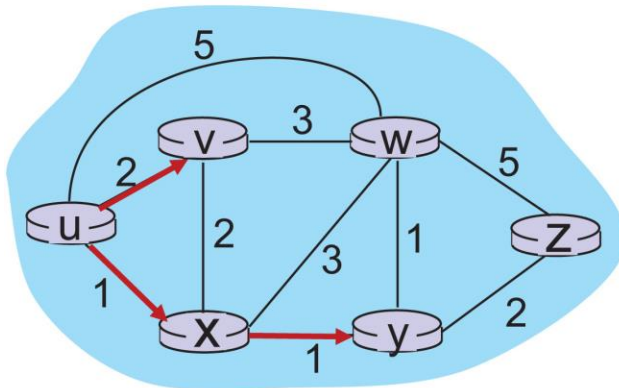
8 Loop

9 find a not in N' such that $D(a)$ is a minimum

10 add a to N'

Dijkstra's Algorithm: An Example (7 of 13)

Step	N'	v D(v),p(v)	w D(w),p(w)	x D(x),p(x)	y D(y),p(y)	z D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2	uxy	2,u	3,y			4,y
3	uxyv		3,y			4,y
4						
5						



8 Loop

9 find a not in N' such that $D(a)$ is a minimum

10 add a to N'

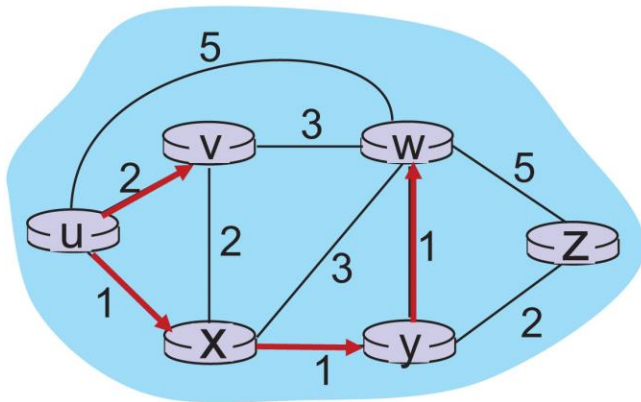
11 update $D(b)$ for all b adjacent to a and not in N' :

$$D(b) = \min (D(b), D(a) + c_{a,b})$$

$$D(w) = \min (D(w), D(v) + c_{v,w}) = \min (3, 2+3) = 3$$

Dijkstra's Algorithm: An Example (8 of 13)

Step	N'	$D(v), p(v)$	$D(w), p(w)$	$D(x), p(x)$	$D(y), p(y)$	$D(z), p(z)$
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x	2,x	∞	∞
2	uxy	2,u	3,y		4,y	
3	uxyv		3,y		4,y	
4	uxyvw					
5						



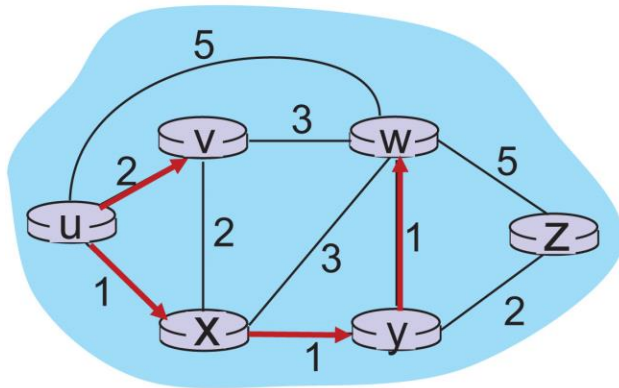
8 Loop

9 find a not in N' such that $D(a)$ is a minimum

10 add a to N'

Dijkstra's Algorithm: An Example (9 of 13)

Step	N'	v D(v),p(v)	w D(w),p(w)	x D(x),p(x)	y D(y),p(y)	z D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2	uxy	2,u	3,y			4,y
3	uxyv		3,y			4,y
4	uxyvw					4,y
5						



8 Loop

9 find a not in N' such that $D(a)$ is a minimum

10 add a to N'

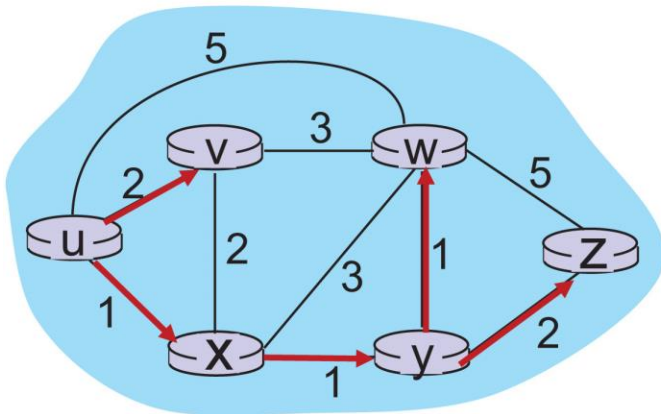
11 update $D(b)$ for all b adjacent to a and not in N' :

$$D(b) = \min (D(b), D(a) + c_{a,b})$$

$$D(z) = \min (D(z), D(w) + c_{w,z}) = \min (4, 3+5) = 4$$

Dijkstra's Algorithm: An Example (10 of 13)

Step	N'	$D(v), p(v)$	$D(w), p(w)$	$D(x), p(x)$	$D(y), p(y)$	$D(z), p(z)$
0	u	2, u	5, u	1, u	∞	∞
1	ux	2, u	4, x		2, x	∞
2	uxy	2, u	3, y			4, y
3	uxyv		3, y			4, y
4	uxyvw					4, y
5	uxyvwz					

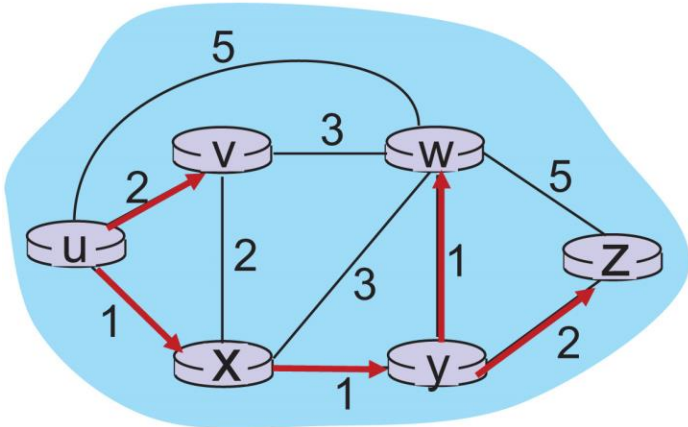


8 Loop

- 9 find a not in N' such that $D(a)$ is a minimum
- 10 add a to N'

Dijkstra's Algorithm: An Example (11 of 13)

		V	W	X	Y	Z
Step	N'	D(v),p(v)	D(w),p(w)	D(x),p(x)	D(y),p(y)	D(z),p(z)
0	u	2,u	5,u	1,u	∞	∞
1	ux	2,u	4,x		2,x	∞
2	uxy	2,u	3,y			4,y
3	uxyv		3,y			4,y
4	uxyvw					4,y
5	uxyvwz					



8 Loop

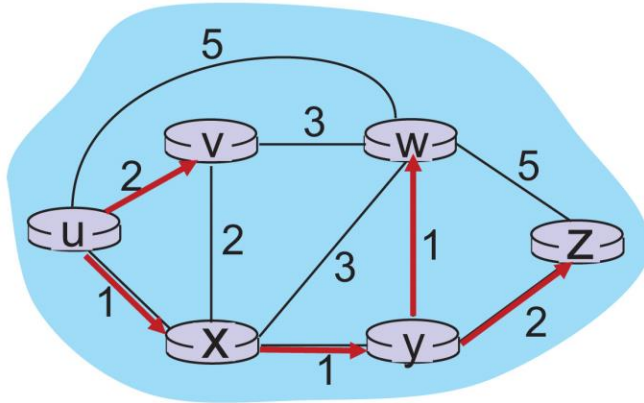
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- 10 add a to N'
- 11 update $D(b)$ for all b adjacent to a and not in N' :

$$D(b) = \min (D(b), D(a) + c_{a,b})$$

Dijkstra's Algorithm: An Example (12 of 13)

Step	N'	$D(v), p(v)$	$D(w), p(w)$	$D(x), p(x)$	$D(y), p(y)$	$D(z), p(z)$
0	u	2, u	5, u	1, u	∞	∞
1	ux	2, u	4, x	2, x	∞	∞
2	uxy	2, u	3, y	4, y	4, y	∞
3	uxyv	2, u	3, y	4, y	4, y	∞
4	uxyvw	2, u	3, y	4, y	4, y	∞
5	uxyvwz	2, u	3, y	4, y	4, y	∞

$D(w), p(w)$
 5, u
 4, x
 3, y
 3, y



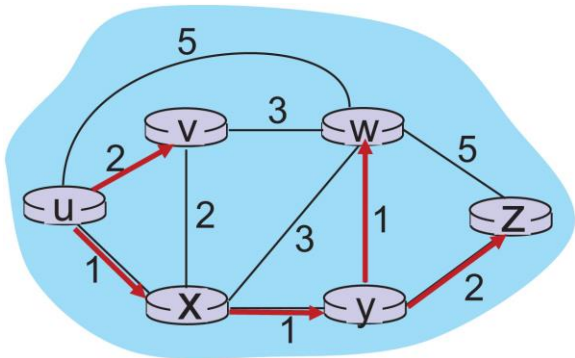
Initialization (step 0): For all a : if a adjacent to then $D(a) = c_{u,a}$

find a not in N' such that $D(a)$ is a minimum
 add a to N'

update $D(b)$ for all b adjacent to a and not in N' :

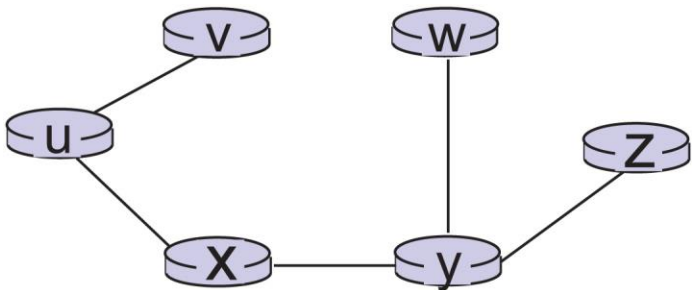
$$D(b) = \min(D(b), D(a) + c_{a,b})$$

Dijkstra's Algorithm: An Example (13 of 13)



$D(w),p(w)$
5,u
4,x
3,y
3,y

resulting least-cost-path tree from u:



resulting forwarding table in u:

destination	outgoing link
v	(u,v)
x	(u,x)
y	(u,x)
w	(u,x)
x	(u,x)

— route from u to v directly

} route from u to all other destinations via x